

COMP101

Introduction to Programming

2025-26

Lab-10

2-d arrays

Objectives: Tasks only – no solutions – refer to lecture notes

Ex-01: create list statically, o/p selected elements

Ex-02: create list by generator, sum array, rows and columns

1-d vs 2-d

2-d list has data as a 'list of lists'

```
meal_list= [ [soup, salad], [fish, chips, peas, beans], [cake, jelly] ]
```

row index: [0] [1] [2]

Outer square brackets define this is a list: `meal_list = [ ]`

Inner brackets define the rows:

there are three rows in this 2-d list

```
meal_list = [ [], [], [] ]
```

Reference in 'row-column order' – the rows come first

Dimension is determined by the row with the most values in it

meal_list[1] has four values: Hence dimension = 3 x 4 (3 rows, 4 columns)

1-d vs 2-d

meal_list= [[soup, salad], [fish, chips, peas, beans], [cake, jelly]]

dimension = 3 x 4 because meal_list[1] has four values in it

‘null’ means empty

soup	salad	null	null
fish	chips	peas	beans
cake	jelly	null	null

Show how the 12 elements are ‘populated’

- ‘1’ means there is a value in it
- ‘0’ is a ‘null element’ – there is nothing in it

1	1	0	0
1	1	1	1
1	1	0	0

Exercise-01: No solution provided

Program the `meal_list` and print out selected elements from it as selected by the user.

This would constitute a customer's order from a set menu

Create list automatically: null by Generator

```
num_of_rows = 4
```

```
num_of_cols = 4
```

Best method for very large lists

```
list_gen = [ [0] * num_of_cols for i in range(num_of_rows) ]
```

```
print (list_gen)
```

```
[[0, 0, 0, 0], [0, 0, 0, 0], [0, 0, 0, 0], [0, 0, 0, 0]]
```

Create a list of 4 rows with 4 elements in each row

In the generator, each element is set to 0

Each element is an 'iterator' and created independently from the others.

[0] * item is constructed row-times – no reference occurs

Exercise-02: No solution provided

- 2.1 Create a unary list using a generator – dimension of your choice
- 2.2 Sum all the values in the array
- 2.3 Sum the total value of each individual row
- 2.4 Sum the total of each individual column